Prateek Anand

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My research interests are in developing novel AI/ML computational methods. Broadly, I am invested in deep generative models, traditional machine learning, and statistical approaches for understanding and generating biological data.

EDUCATION

UCLA | *Ph.D., Computer Science* | Advisor: Dr. Sriram Sankararaman **2024 - 2029**
- Recent work, GPC (deep generative model for genetic data): *PLOS Genetics*; **Top 10% Oral Spotlight**, *ICLR Gen² Workshop*

UCLA | *M.S., Computer Science* **2024 - 2026**

UCLA | *B.S., Computer Science* **2020 - 2024**

PROFESSIONAL EXPERIENCE

UCLA Computer Science | Sriram Lab **Jun 2022 - Present**

Graduate Student Researcher
- Developing discrete diffusion models and improved guidance methods for cell-type-specific enhancer design (under review)
- Building deep generative models of genetic variation for synthetic data and imputation
- Developing ensemble machine learning approaches for imputing rare variants across diverse ancestries
- Creating scalable machine learning/statistical software to understand human genetic architecture

Research Intern (Bruins in Genomics Research Program)
- Evaluated a symbolic regression-based explanation method for complex black box machine learning models

Stanford University School of Medicine | Curtis Lab **Jun 2023 - Jan 2024**

Research Intern (Canary CREST Research Program for Early Cancer Detection)
- Computational modeling and inference, refined model built using approximate Bayesian computation to detect blood cancer early

UCLA Jonsson Comprehensive Cancer Center | Boutros Lab **Jan 2021 - Apr 2022**

Research Assistant
- Updated, maintained, and evaluated CI/CD HATCHet pipeline for use on lab cluster compute server

PUBLICATIONS

** denotes equal contribution*

- **Prateek Anand**, Anji Liu, Meihua Dang, Boyang Fu, Xinzhu Wei, Guy Van den Broeck*, Sriram Sankararaman*. GPC: An expressive and tractable deep generative model for genetic variation data, *PLOS Genetics*, (2026). Also a **Top 10% Oral Spotlight** at the *ICLR Gen² Workshop* (2026).
- Yash Patel*, Chenghao Zhu*, Takafumi N. Yamaguchi*, ..., **Prateek Anand**, ..., Paul C. Boutros. Metapipeline-DNA: A comprehensive germline and somatic genomics Nextflow pipeline, *Cell Reports Methods*, (2026).
- Boyang Fu*, Ali Pazokitoroudi*, ..., **Prateek Anand**, ..., Sriram Sankararaman. A biobank-scale test of marginal epistasis reveals genome-wide signals of polygenic interaction effects, *Nature Genetics*, (2025).
- Boyang Fu*, **Prateek Anand***, Aakarsh Anand*, Joel Mefford, Sriram Sankararaman. A scalable adaptive quadratic kernel method for interpretable epistasis analysis in complex traits, *Genome Research*, (2024). Also presented at *RECOMB* (2024).

PREPRINTS

- Moonseong Jeong, ..., **Prateek Anand**, ..., Sriram Sankararaman. Genome-wide LD scores enable robust heritability and genetic correlation estimation from summary statistics across genetic ancestries, *bioRxiv*, (2026).
- Zhengtong Liu, Boyang Fu, Moonseong Jeong, **Prateek Anand**, ..., Sriram Sankararaman. Comprehensive gene heritability estimation reveals the genetic architecture of rare coding variants underlying complex traits, *bioRxiv*, (2025).

PRESENTATIONS

- **Prateek Anand**, ..., Guy Van den Broeck*, Sriram Sankararaman*. Deep generative model of genetic variation data improves imputation accuracy in private populations, *American Society of Human Genetics*, (2025).

TEACHING

- **Machine Learning Applications in Genetics (UCLA CS 124):** Teaching Assistant (Spring 2026)
- **Machine Learning Applications in Biomedicine (UCLA DSB 205):** Teaching Assistant (Winter 2026)
- **Introduction to Machine Learning (UCLA CS 146):** Teaching Assistant (Winter 2025); Reader (Winter 2023, Winter 2024)

SKILLS

- **Programming & Tools**: Python, C++, PyTorch, Scikit-learn, NumPy, Pandas, Git, Cluster Computing
- **Expertise**: Deep Learning, Generative Models, Diffusion Models, Machine Learning, Statistics, Probabilistic Models

AWARDS

- **Google PhD Fellowship Nominee:** One of four nominated by UCLA (2026)
- **Warren Alpert Computational Biology and AI Fellow:** Graduate Training Fellowship (2024)
- **NCI Scholarship:** Canary CREST Program (2023); **NSF REU Scholarship:** Bruins in Genomics Program (2022)
- **Andy Grove Intel Scholarship:** Academic Excellence (2020); **UCLA Dean's Honors List** (2020 - 2024)